

Problem 1

(i) $7^{-1} \equiv 8 \pmod{11}$

(ii) $\gcd(26, 10) = 2 \Rightarrow 10^{-1} \pmod{26}$ does not exist.

(iii) $11^{-1} \equiv 17 \pmod{31}$

(iv) $23 \times 4 = 92 \equiv -1 \pmod{31} \Rightarrow 23^{-1} \equiv -4 \equiv 27 \pmod{31}$

(v) $237 = 3 \times 79$. $91 \equiv 1 \pmod{3} \Rightarrow 91^{-1} \equiv 3k+1 \pmod{3}$
 $91 \equiv 12 \pmod{79} \Rightarrow 91^{-1} \equiv 79l+33 \pmod{79}$ $\searrow \rightarrow 112$
 $\Rightarrow 91^{-1} \equiv 112 \pmod{237}$

Problem 2. Suppose Adam moves first and Bob moves second.

Claim: Bob has a winning strategy when $n=4k$. ($n=00$: Bob; $n=2026$, Adam)

Adam has a winning strategy when $n=4k+1, 4k+2, 4k+3$.

Proof: (a) $n=4k$. At each step, if Adam chooses 1, the Bob chooses 3;
if Adam chooses 2, the Bob chooses 2;
if Adam chooses 3, the Bob chooses 1.

Observation: ① After Adam's operation, the existing number is not divisible by 4.

② Bob can reach $4k$ for all $k \geq 1$.

Conclusion: Bob has a winning strategy.

(b) $n=4k+1$. Adam chooses 1 at the first step and 'steals' the above strategy for Bob.

(c) $n=4k+2, 4k+3$. Same as (b).

Problem 3. (i) $\gcd(a, n) = 1 \Rightarrow \exists A, Aa \equiv 1 \pmod{n}$
 $\Rightarrow (AB) \cdot (ab) \equiv 1 \pmod{n}$
 $\gcd(b, n) = 1 \Rightarrow \exists B, Bb \equiv 1 \pmod{n}$
 $\downarrow$
 $\gcd(ab, n) = 1.$

Existence:

(ii) $n_1, \dots, n_k$ are pairwise coprime $\Leftrightarrow \gcd(n_i, n_j) = 1$ for any $1 \leq i < j \leq k$.

By (i) and induction, $\gcd(n_1, n_2 n_3 \dots n_k) = 1$.

Hence, there exists N_1 such that $N_1 n_2 n_3 \dots n_k \equiv 1 \pmod{n_1}$

Let $M_1 = N_1 n_2 \dots n_k$. Define $M_2, \dots, M_k$ similarly.

It follows that $M_i \equiv 1 \pmod{n_i}$ and $M_i \equiv 0 \pmod{n_j}$ for $j \neq i$.

Hence $x = \sum_{i=1}^k a_i M_i$ is a solution of the system of congruences.

Uniqueness:

Suppose we have two solutions x_1 & x_2 . Then

$$x_1 - x_2 \equiv 0 \pmod{n_i} \text{ for all } 1 \leq i \leq k.$$

Since $n_1, n_2, \dots, n_k$ are pairwise coprime,

$$x_1 - x_2 \equiv 0 \pmod{n_1 n_2 \dots n_k}.$$

Hence, the solution is unique modulo $n_1 n_2 \dots n_k$.

Problem 4

(i) $\gcd(3, 12) = 3$. $3 \nmid 1 \Rightarrow$ no solution.

(ii) $x \equiv 4 \pmod{11}$ (iii) $x \equiv 6 \pmod{7}$

(iv) $\gcd(15, 18) = 3$. $3 \nmid 5 \Rightarrow$ no solution.

(v) $x \equiv 7 \pmod{20}$

Problem 5. (i) $x = 4 + 24k \equiv 4 + 2k \pmod{11}$. $x \equiv 7 \pmod{11} \Rightarrow k \equiv 7 \pmod{11}$

$$\Rightarrow k = 11l + 7, \quad x = 172 + 264l.$$

(iii)

4, 12, 20, 28, 68, 108, 148

+8, +8, +8, +40, +40, +40

4, 28, 148

$$4 \equiv 4 \pmod{8}, \quad 28 \equiv 3 \pmod{5}, \quad 148 \equiv 1 \pmod{7}$$

Conclusion: $x \equiv 148 \pmod{280}$.